Practical : 9	Implement prim's algorithm.
Date : 19/10/2026	

Aim: To implement Prim's Algorithm to find the Minimum Spanning Tree (MST) of a connected, weighted, undirected graph**.**

Program:

```
vertices = 5

graph = [
    [0, 2, 0, 6, 0],
    [2, 0, 3, 8, 5],
    [0, 3, 0, 0, 7],
    [6, 8, 0, 0, 9],
    [0, 5, 7, 9, 0]
]

visited = [False] * vertices
visited[0] = True

total_cost = 0
edges = 0

print("Edges in Minimum Spanning Tree:")

while edges < vertices - 1:
    minimum = float('inf')
    start = -1
    end = -1

    for i in range(vertices):
        if visited[i]:
            for j in range(vertices):
                if not visited[j] and graph[i][j] != 0:
                 if graph[i][j] < minimum:
                    minimum = graph[i][j]
```

```
                start = i
                end = j

    if end == -1:
        print("Graph is not connected.")
        break

    print(start, "-", end, ":", minimum)

    visited[end] = True
    total_cost += minimum
    edges += 1

if edges == vertices - 1:
    print("Total cost of MST:", total_cost)
```

Output:-

```
Microsoft Windows [Version 10.0.26200.9457]
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C:\Users\Vishnu\OneDrive>python prims.py
Edges in Minimum Spanning Tree:
0 - 1 : 2
1 - 2 : 3
1 - 4 : 5
0 - 3 : 6
Total cost of MST: 16
```

Analysis:

Time Complexity: $O(V^2)$
Space Complexity: $O(V^2)$
Where V is the number of vertices. The adjacency matrix requires $O(V^2)$ space.

Conclusion

Prim's Algorithm was implemented successfully using the greedy approach to find the Minimum Spanning Tree with a minimum total cost of 16.